Huawei Access Node Failure SOP - BAS-ACC-53 - Basavanagudi POP

Node: BAS-ACC-53 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1970

Description

This Standard Operating Procedure (SOP) outlines the steps to investigate and resolve a warning-level optical transport failure occurring on the Huawei Access Node (BAS-ACC-53) at the Basavanagudi POP. The failure is characterized by the appearance of alarm code ALARM_1970, indicating degradation of the SFP/XFP, fiber cuts, or loss of receive/transmit power. This failure domain is unique to the Access role on Huawei equipment, where the Optical Transport Network (OTN) protocol, utilizing the G.709 standard, governs the transmission and reception of optical signals over fiber links. The OTN protocol relies on a complex interplay of lasers, photodetectors, and optical amplifiers to maintain a stable light signal. Any degradation or interruption in these components can result in the observed alarm condition. Furthermore, the Huawei Access Node's software, running on the Huawei's proprietary OSN 8800 platform, plays a critical role in monitoring and controlling the OTN protocol. The node's control plane, utilizing the IETF-defined OTN protocol, communicates with the data plane to maintain the integrity of the optical signal. In the event of a failure, the control plane detects anomalies and triggers alarms, including ALARM_1970, to alert network operators. This SOP outlines the necessary steps to diagnose and rectify the issue, ensuring minimal service disruption and maintaining the integrity of the optical transport network.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and all necessary equipment and personnel are on standby.

- Check the status of redundant links and ensure they are functioning correctly.

- Confirm that remote hands are dispatched to the POP site.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP module	High
Fiber cut or damage	Medium
Rx/Tx power loss due to optical amplifier failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport alarm info, Expected Output: ALARM_1970: SFP/XFP degradation/fiber cut/Rx/Tx power loss, Description: This command displays the current alarm information, including the alarm code, description, and affected interface. It helps identify the root cause of the failure.
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```
Command: display optical-transport interface info, Expected Output: Interface status: Down/Up, SFP/XFP type, optical power level, Description: This command provides detailed information about the optical interface, including its status, SFP/XFP type, and optical power level. It helps verify if the issue is related to the interface or the optical signal.

Command: display optical-transport tx/rx power info, Expected Output: Tx/Rx power level, Optical Signal-to-Noise Ratio (OSNR), Description: This command displays the transmit and receive power levels, as well as the OSNR, which helps determine if the issue is related to the optical signal quality.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport sfp-xfp, Description: Replace the degraded SFP/XFP module with a new one to restore the optical signal integrity.

Step 2: Step: Verify fiber integrity and repair if necessary, Command: verify fiber integrity, Description: Check the fiber for any damage or cuts and repair it if necessary to restore the optical signal.

Step 3: Step: Reset the optical amplifier, Command: reset optical-amplifier, Description: Reset the optical amplifier to restore the Rx/Tx power level and ensure proper signal transmission.

Post-Check Verification

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Command: display optical-transport alarm info, Expected Output: No alarm, Description: Verify that the alarm is cleared and the optical signal is stable.

Command: display optical-transport interface info, Expected Output: Interface status: Up, SFP/XFP type, optical power level, Description: Verify that the optical interface is up and the SFP/XFP module is functioning correctly.

Command: display optical-transport tx/rx power info, Expected Output: Tx/Rx power level, OSNR within normal range, Description: Verify that the Tx/Rx power levels and OSNR are within the normal range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace optical-transport sfp-xfp original, Description: Restore the original SFP/XFP module to revert the changes made during the resolution process.

- Step: Re-verify fiber integrity, Command: verify fiber integrity, Description: Re-verify the fiber integrity to ensure that the repair was successful.

- Step: Re-reset the optical amplifier, Command: reset optical-amplifier, Description: Re-reset the optical amplifier to restore the original settings.